THESIS TITLE

A PREPRINT

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ABSTRACT

Some super cool abstract of all my findings

1 Introduction

A central goal of mechanistic interpretability is to decompose a neural network's parameters into a small number of meaningful units, atoms, each of which corresponds to something the model is doing. The hope is that a network's computation, however dense and distributed in its raw form, can be expressed as a sum of simpler mechanisms that admit human-readable descriptions.

A recent line of work shifts the object of decomposition from activations to parameters, decomposing a network's weights into a sum of simpler components, each taken to correspond to a concrete atom of computation. Methods guided by the principle of minimum description length aim to recover components that are faithful to the original weights, minimal in the number required per input, and individually as simple as possible. Stochastic Parameter Decomposition (SPD) [Bus+25] is one such method, decomposing networks into rank-one subcomponents and learning a per-component importance function that estimates each atom's causal contribution to a given input.

The Open Problems in Mechanistic Interpretability programme [Sha+25] identifies the question of how best to carve a network at its joints as a central open problem of the field. Whether a parameter decomposition recovers the computational primitives the model actually uses depends on what the ground-truth decomposition is taken to be, and the carving question is the question of which units to take as ground truth. For models with independent features the answer is comparatively unambiguous: one atom per feature. The question becomes substantially harder when the features themselves are dependent, a regime that is empirically attested in real models. Engels et al. [Eng+24] develop a rigorous notion of irreducible multi-dimensional features, defined as features whose joint distribution can be split neither into independent components nor into disjoint mixture components, and show that language models represent inherently cyclical concepts, including days of the week and months of the year, as two-dimensional irreducible features whose individual coordinates are not independent units but components of a single object the model manipulates together. Networks that represent such features admit multiple decompositions that are equally faithful to the weights but differ in the level at which features are separated. A network that computes the volume of a cube could equally faithfully be decomposed into separate components that each compute one of the side lengths, or into a single component that computes the volume directly. The first is a feature-level decomposition: one atom per individual input direction, exposing the internal structure of the computation but leaving the grouping of features implicit, recoverable only by inspecting which atoms co-activate. The second is a structural-level decomposition: one atom per irreducible structure, exposing the grouping directly but treating the internal computation as a single unit. Each level loses something the other preserves.

The present work extends SPD by examining its behaviour on dependent feature pairs and larger structures, configurations in which the multi-dimensional superposition hypothesis posits a structural-level unit. Here SPD is shown to recover both feature-level and structural-level decompositions, and the conditions distinguishing the two outcomes are characterised. Building on this, we introduce softmax-and-attention competition between subcomponents, making the importance calculation relational, and show that the extension allows SPD to recover decompositions that respect the multi-dimensional superposition hypothesis. We additionally introduce the Toy Model of Geometry, a small model that computes the volume of a simplex, to examine how this method performs on a more complex structure.

2 Old Introduction - may be useful for related work.

Neural networks continue to achieve remarkable performance across domains ranging from medical diagnosis to autonomous systems. This increase in performance has, however, come with a corresponding increase in complexity, making these systems less interpretable to those who build and deploy them. As neural networks enter high-stakes environments, where errors carry legal, financial, or human cost, interpretability becomes a necessity. The concern is not merely technical but ethical: a fundamental principle of accountability holds that the creator of a tool can only be held responsible for its outcomes to the extent that those outcomes could have been reasonably foreseen. A system whose internal reasoning cannot be understood is, by definition, one whose failures cannot be anticipated.

Mechanistic interpretability aims to reverse-engineer the internal computations of neural networks into their simplest forms, such that they can be expressed as humanly interpretable algorithms.
Many of the current popular methods within this space, such as sparse autoencoders [Cun+23], circuit analysis, and probing, primarily operate in activation space. However, each of these methods comes with its own set of limitations. Sparse autoencoders treat features as independent directions and discard the geometric structure between them, despite growing evidence that this structure carries meaningful information (Leask et al., 2025; Mendel, 2024; Engels et al., 2024). Moreover, sparse autoencoders are trained to reconstruct activations and do not to explicitly identify the computational units that produced them. [Lea+25; Cha+25] The resulting decomposition may be functionally equivalent at a given layer, while not resembling the mechanisms that the network employs. Circuit-level methods face the deeper issue that neural networks are densely connected, meaning any given output can be equally well attributed to a multitude of causal pathways, leaving the choice of circuit underdetermined.

A more recent line of work shifts the object of decomposition from activations to parameters. More concretely, these methods aim to decompose a network's weights into a sum of simpler components, each of which corresponds to a concrete "mechanism". This reframing sidesteps several of the issues above: mechanisms are no longer constrained to specific architectural components, such as neurons or attention heads, and can instead be distributed across multiple architectural components. Moreover, they decompose the network into functional units rather than representational directions, which may further enhance their interpretability.
Recently proposed methods guided by the principle of minimum description length have shown promising results in this direction, aiming to decompose a network into parameter components that are faithful to the original weights, minimal in the number required per input, and individually as simple as possible.
One such method, Stochastic Parameter Decomposition (Bushnaq et al., 2025), decomposes networks into rank-one subcomponents. Since not all components are relevant for every input, the method learns a function that estimates each component's causal importance for a given input. However, each component's importance is calculated independently, relying only on its own activation. As noted in the original work, this is likely too restrictive for non-toy models, as it prevents each component's importance from being informed by the state of other components. This work replaces the independent importance functions with a graph neural network that determines each component's importance conditioned on its relationships to all other components, while preserving explainability through its generalised additive structure. Additionally the use of an additive importance function, generalizes the decomposition to k-dimensional components. This generalizsation is motivated by the growing evidence, that transformer models, learn features which occupy irreducible multidimensional subspaces.

[Bra+25]

3 Related work

-Open problems in mechanistic interpretability - SPD - SPD on transformers - APD - Engels et al. -Activation space decompositions - Circuits, SAE, Linear representation hypothesis Superposition in general? Strong feature hypothesis

4 Method

4.1 Background on APD and SPD

The initial inspiration for shifting the focus from activations to parameters came from Attribution-based Parameter Decomposition (APD) [Bra+25]. APD decomposes the weights of a network into a set of parameter components: vectors in parameter space that satisfy three desirable properties.

Let $f(x, W)$ be a neural network with weight matrices $W = \{W^1, \ldots, W^L\}$. A set of parameter components should satisfy the following properties:

- **Faithfulness**: The parameter components should sum to the parameters of the original network.
- **Minimality**: As few parameter components as possible should be required to replicate the network's output on any given input.
- **Simplicity**: Each component should span as few weight matrices and as few ranks as possible.

If a set of parameter components satisfies these properties, they are said to comprise the network's mechanisms [Bra+25].

However, APD suffers from several practical limitations. Each parameter component is a full vector in parameter space $\mathbb{R}^{|\theta|}$, where $|\theta|$ is the total number of parameters in the network, thus increasing the memory cost to $O(\theta \cdot C)$, where C is the number of components, making the method computationally expensive to scale to larger models. APD is also highly sensitive to its top-k hyperparameter, which specifies the expected number of active components per input and must be chosen in advance. Lastly, APD relies on the magnitude of the gradient as a proxy for the causal importance of each component. These gradients have been shown to be poor approximations of the true importance. (cite the papers here).

Stochastic parameter decomposition

Stochastic Parameter Decomposition (SPD) was introduced as a solution to the limitations of APD. SPD decomposes each weight matrix into a sum of rank-one matrices, called subcomponents:

$$W^l \approx \sum_{c=1}^{C} \vec{u}_c^l \vec{v}_c^{lT}, \quad \text{where} \quad \vec{u} \in \mathbb{R}^{d_{out}}, \vec{v} \in \mathbb{R}^{d_{in}}$$

The number of subcomponents C is a hyperparameter that may be chosen to be larger than the rank of W^l enabling the method to learn features in superposition [BBS25].

These subcomponents are then trained to be faithful to the original weights, to have as few active subcomponents as possible for any given input, and lastly to reconstruct the original network's output.

4.1.1 Faithfulness

4.1.2 Stochastic reconstruction

4.1.3 Minimality

4.1.4 Limitations

These components are later clustered manually.... -¿ GNaN solves Components are rank 1, many features are multidimensional, dependent components are trained independently

4.2 Irreducibility and the natural unit of decomposition

The standard formulation of superposition treats features as one-dimensional directions in activation space, with each feature contributing a scalar activation along its direction.

The work of engels et al. provide empirical evidence that the strong feature hypothesis is too restrictive, and that transformer architectures organise cyclical concepts, such as days of the week and months of the year, as two-dimensional structures, that are not composed of independent one-dimensional features and instead are irreducibly multi-dimensional.

To accomodate this, the paper provides a formal definition of irreducibility and a corresponding refinement of the superposition hypothesis.

Irreducibility. A feature $f \in \mathbb{R}^{d_f}$ is reducible into features a and b if there exists an affine transformation $f \mapsto Rf + c \equiv (a, b)$ with R orthonormal, such that the joint distribution $p(a, b)$ satisfies one of two conditions:

- **Separability**: $p(a, b) = p(a)\, p(b)$
- **Mixture of disjoint distributions**: $p(a, b) = w\, p(a)\, \delta(b) + (1 - w)\, p'(a, b), \quad w \in (0, 1)$

where δ is the Dirac delta distribution. In other words, a feature is reducible when its components are statistically independent, or when the joint distribution decomposes into a regime where one component is exactly zero and the other varies on its own.

A feature whose joint distribution admits neither of these properties, carries information only when interpreted together, and any decomposition into smaller units would lose information the original feature encodes.

With this definition, the superposition hypothesis is refined as follows:

Every hidden state $x_{i,l}(t)$ can be written as a sum of feature contributions

$$x_{i,l}(t) = \sum_i V_i \, f_i(t) \tag{1}$$

where each $f_i(t) \in \mathbb{R}^{d_{f_i}}$ is an irreducible feature with internal dimensionality $d_{f_i} \geq 1$, each $V_i \in \mathbb{R}^{d \times d_{f_i}}$ is a projection matrix embedding the feature into a d_{f_i}-dimensional subspace of hidden space, and the projection matrices are pairwise δ-orthogonal: $\|V_i^\top V_j\| < \delta$ for $i \neq j$.

The multi-dimensional hypothesis makes two structural commitments about the model's representation, that will be explored throughout this work. Firstly, the projection matrices V_i are pairwise δ-orthogonal, with the column spaces of different features approximately disjoint in hidden space. Secondly, each f_i is irreducible, meaning the coordinates within a single feature covary according to the feature's joint distribution rather than being independent components that happen to share a subspace. [1]

The definition of $V_i \in \mathbb{R}^{d \times d_{f_i}}$ does not require its columns to be linearly independent: d_{f_i} specifies the formal dimensionality of the irreducible feature, but $\text{rank}(V_i) \leq d_{f_i}$ is what the model actually uses. The two coincide when the joint distribution of the feature's coordinates is non-degenerate enough that compression would lose information; they come apart when the distribution permits a lower-rank representation. Two perfectly correlated coordinates with $f_i^{(1)}(t) = f_i^{(2)}(t)$ for all t, for instance, can be represented through a V_i with both columns equal to a single direction $v \in \mathbb{R}^d$:

$$V_i \, f_i = v \cdot f_i^{(1)} + v \cdot f_i^{(2)} = 2v \cdot f_i^{(1)} \tag{2}$$

Here V_i is formally $d \times 2$ but its column space is one-dimensional, and the feature is irreducibly two-dimensional in its input distribution but represented through a rank-deficient V_i without loss of reconstructive information. We will show empirical evidence of this phenomenon in section **??**.

The formal dimensionality d_{f_i} thus acts as an upper bound on the rank a faithful representation requires, with the actual rank determined by how the model has chosen to encode the feature.

Implications for parameter decomposition. This present work takes the irreducible feature as the atomic unit of decomposition, and explores how to extend SPD to recover these features both when they can be collapsed into a single rank-one component and what happens when they cannot. This definition of irreducibility formalises the structural-level of decomposition described in the introduction. The decomposition in the dependent feature regime should aim to recover these irreducible features, and not split them into independent components or collapse them into larger units.

4.3 Rank-k subcomponents and relational importance

The framework of Engels et al. described in section 4.2 identifies that these irreducible features are the desired units of decomposition, with internal dimensionality d_{f_i}. SPD's rank-one parameterisation can contain such a feature as a single atom only when its effective representational rank is one. [2] Otherwise the decomposition must split these features across multiple subcomponents.

To accommodate this, each pair of vectors in SPD's parameterisation is replaced with a pair of matrices:

$$W^l \approx \sum_{c=1}^{C} U_c^l {V_c^l}^\top, \qquad U_c^l \in \mathbb{R}^{d_{\text{out}} \times k}, \quad V_c^l \in \mathbb{R}^{d_{\text{in}} \times k} \tag{3}$$

Each subcomponent $U_c^l {V_c^l}^\top \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ is a matrix of rank at most k, and setting $k = 1$ recovers the original SPD formulation.

This definition allows each subcomponent to span up to k rank-one factors but does not require it to do so; mechanisms for encouraging atoms to use less than their full capacity are discussed in Section **??**.

[1] Does this not also help my case of softmax, we train the importance to maximize the importance, some old paper i read meant that it increased the intra cluster similarity, and decreased the inter cluster, We should check this over training and see if it happens!

[2] We Say effective rank, as if the model compresses a 2D direction into a line, then we can actually represnet it

The extension to the importance function follows naturally from the change in parameterisation. The inner activation used to estimate each subcomponent's importance is now a k-dimensional vector:

$$h_c^l(x) = {V_c^l}^\top \vec{a}^l(x) \in \mathbb{R}^k \tag{4}$$

where $\vec{a}^l(x)$ is the activation of layer l for input x.

Lastly we replace the independent importance function with a relational one instead of the original independent one, where each components importance was derived from its own activation, independent of all other subcomponents. The justification for this is shown in Section **??**.

We aim to motivate the importance through a probabilistic interpretation, under which importance becomes the posterior probability that a given subcomponent is responsible for the output, conditioned on the activations of all subcomponents in the same layer:

$$\alpha_c^l(x) = p\big(c \mid h_c^l(x), \{h_{c'}^l(x)\}_{c' \neq c}\big) \tag{5}$$

[3] Importance becomes the probability that subcomponent c owns the input, conditioned on the activations of all subcomponents.

The posterior is realised through a softmax over subcomponents,

$$\alpha_c^l(x) = \frac{\exp s_c^l(x)}{\sum_{c'=1}^{C} \exp s_{c'}^l(x)} \qquad s_c^l(x) = f_{\text{self}}(h_c^l(x)) + f_{\text{agg}}\big(\{h_{c'}^l(x)\}_{c' \neq c}\big) \tag{6}$$

The f_{self} maintains the original signal from c's own activation, while f_{agg} acts as a corrective term that contextualises c against other subcomponents through an attention mechanism.

A significant advantage of the reformulation is that softmax naturally enforces a probabilistic distribution over subcomponents. The importance becomes more interpretable: a subcomponent is predominantly responsible for the reconstruction when its allocation dominates. Softmax introduces a structural pressure that encourages each subcomponent to maximise the probability that a feature belongs to it, motivating the optimiser to group dependent features within the same subcomponent.

During the experiments in SPD, this structural pressure was not necessary, as experiment features were indepndent and subcomponents were kept at rank 1.

The rank-k extension lifts this restriction by giving each subcomponent capacity to host up to k rank-one factors. With this added capacity, the question of how dependent features distribute across subcomponents becomes consequential: a mechanism is required to encourage their co-location within a single subcomponent rather than their dispersion across many. The minimality loss originally proposed for SPD takes the form

$$\mathcal{L}_{\text{minimality}}^{\text{SPD}} = \sum_{l=1}^{L} \sum_{c=1}^{C} |g_c^l(x)|^p, \qquad p > 0, \tag{7}$$

penalising a power of the importance magnitudes. The pressure this loss exerts on allocation shape depends on p and is limited in either direction: for $p < 1$ the loss weakly favours concentration, and for $p > 1$ it favours distribution.

Under the softmax reformulation, the minimality loss becomes degenerate: The sum is always one, regardless of how the importance is distributed. We replace it with an entropy-based minimality term that penalises the shape of the allocation:

$$\mathcal{L}_{\text{minimality}} = -\sum_{l=1}^{L} \sum_{c=1}^{C} \alpha_c^l(x) \log \alpha_c^l(x) \tag{8}$$

Entropy is minimised at one-hot allocations, where a single subcomponent has all the importance, and increases monotonically as the allocation flattens.

This pressure is appropriate when each irreducible feature can be hosted by a single subcomponent of sufficient rank. Two regimes fall outside this assumption. The first is when a feature's effective representational rank exceeds the chosen k, so the feature cannot be hosted as a single unit and the second being if a feature is reducible, and should be split across multiple subcomponents. The former serves as a motivation for the rank-k extension, while the latter must be handled through the relative weighting of the faithfulness loss and the minimality loss. If the faithfulness loss dominates, the model is pressured to split these features to achieve a more faithful reconstruction.

[3] maybe just do all components? Does it matter?

It is noted, that this attention mechanism has computational cost $O(l \cdot C^2)$, where C is the number of subcomponents per layer and l is the number of layers, leading to scalability concerns at larger sizes.

[4].

4.4 Add the part that rank k, is unsifficient in the standard SPD, as it has no structural pressure to group things, and always that orthogonal dparts are split across components, as we only punish multiple parts being active, but we dont mind that they are split across multiple components

4.5 SVD reformulation, capacity and per feature ablation

The competition pressure introduced by the softmax has a theoretical motivation under the multi-dimensional superposition hypothesis. The hypothesis predicts that different irreducible features occupy approximately orthogonal projection subspaces; the softmax-and-attention mechanism implements a training pressure toward this orthogonality at the level of SPD's atoms. Atoms whose V matrices share input directions cannot be cleanly distinguished on inputs that activate those directions, and the softmax penalises ambiguous allocations. The optimiser's response is to push the atoms' V matrices toward disjoint subspaces, recovering the orthogonality structure the hypothesis posits.

Remember to add an experiment that shows if softmax enforce the orthogonality constraint from engels.

4.5.1 Per-dimension ablation and implicit rank selection (Hypothesis: ask lukas)

Following the extension to rank-k subcomponents, and the alternative formulation of the GNAN, the output $g_c^l \in [0,1]^k$ is a per feature importance value, and thus the masking can be applied independently to each feature. The intuition behind this is that if a mechanism captures a 1 dimensional feature, it would only need of the k dimensions to be active.
Likewise if this actually ends up working, k can be considered an upper bound or a so called "capacity" of a mechanism, it also follows nicely with the descriptive length principle found in the appendix of APD.

To determine the effective rank of each subcomponent, the stochastic masking becomes:

$$[m_c^l(x,r)]_j = [g_c^l(x)]_j + (1 - [g_c^l(x)]_j) r_{c,j}^l, \qquad r_{c,j}^l \sim U(0,1)$$

for each j in $1, \ldots, k$
Notably this treats each feature independently, which may not be desirable, as such we could modify it ever so slightly:

$$[m_c^l(x,r)]_j = [g_c^l(x)][g_c^l(x)]_j + (1 - [g_c^l(x)][g_c^l(x)]_j) r_{c,j}^l, \qquad r_{c,j}^l \sim U(0,1)$$

The signal becomes an importance of two factors, if the component is important and both features are important, it remains active, if only one feature is important, the other will be pushed to zero

The masked weight matrix becomes:

$$W'^l(x,r) = \sum_{c=1}^{C} U_c^l \operatorname{diag}\big(\mathbf{m}_c^l(x,r)\big) V_c^{l\top}$$

Where the diagonal matrix is the per-feature mask for subcomponent c.

The importance minimality loss penalises all k dimensions of all subcomponents toward zero:

$$\mathcal{L}_{\text{import}} = \sum_{l=1}^{L} \sum_{c=1}^{C} \sum_{j=1}^{k} \big|[g_c^l(x)]_j\big|^p$$

This follows the same logic by which SPD selects the number of active subcomponents: the minimality loss drives unused components toward zero, and only those that are genuinely required to reconstruct the target model's output survive.

This makes k a soft upper bound rather than a sensitive hyperparameter. In practice, k can be set conservatively large and the effective rank of each subcomponent is learned from the data.

[4]Double check this, flex? attention was also used by someo ther paper on SPD and transformers, but applied to the sequence length.

5 Experiments

5.1 Reproducing SPD on independent features

5.1.1 Sensitivity to hyperparameters

5.2 TMS with tied weights and synced inputs

To examine how SPD and the softmax-and-attention extension handle irreducible features, we reuse the model of superposition introduced in Section 5.1. The architecture is a single-hidden-layer autoencoder with tied weights:

$$f(x) = W^\top \sigma(Wx + b), \qquad W \in \mathbb{R}^{d_{\text{hidden}} \times d_{\text{features}}}, \quad d_{\text{features}} \gg d_{\text{hidden}}$$

The model is trained to reconstruct sparse feature activations under an L_2 loss. The input distribution differs from Section 5.1 in that features are tied into pairs. Concretely, $d_{\text{features}} = 40$, $d_{\text{hidden}} = 10$, and the features are partitioned into 20 pairs $\{f_{2i-1}, f_{2i}\}_{i=1}^{20}$. Each pair is zero with probability $1-p$, and with probability p both features are independently drawn from $\mathrm{Uniform}(\epsilon, 1)$.

By construction, the joint distribution of each pair satisfies neither of the reducibility conditions defined in Section 4.2: the activation of one feature in a pair fully determines the activation of the other, and the joint distribution has no mass where one feature is zero while the other is active. By [Eng+25], each pair is therefore an irreducible two-dimensional feature.

Since $d_{\text{features}} \gg d_{\text{hidden}}$, feature representations are necessarily compressed through superposition. In the learned encoder, this results in near-collinearity within feature pairs and approximate orthogonality across pairs.

Group	Mean	Std Dev	Range
Within-group	0.999	0.010	$[0.997, 1.000]$
Across-group	0.053	0.223	$[-1.000, 0.038]$

Table 1: Cosine similarity between encoder columns $W_{:,i}$ and $W_{:,j}$, partitioned by group membership.

Under the multi-dimensional superposition hypothesis (Section 4.2), the formal dimensionality $d_{fi} = 2$, which provides an upper bound on the rank required to represent each feature. The observed near-collinearity within pairs suggests that this representation essentially collapses to a one-dimensional subspace.

To quantify this, the effective rank of each pair's submatrix, $W_{:,2i-1:2i} \in \mathbb{R}^{d_{\text{hidden}} \times 2}$ is computed using the entropy-based definition of [RV07].

$$r_{\text{eff}}(W_g) = \exp\left(-\sum_{k=1}^{Q} p_k \log p_k \right), \qquad p_k = \frac{\sigma_k^2}{\sum_{j=1}^{Q} \sigma_j^2}$$

where σ_k are the singular values of W_g and $Q = |g|$. The quantity lies in $[1, |g|]$ and equals 1 when the group is exactly rank-one. To gauge the required value of k for the rank-k extension, the maximum effective rank across pairs is examined:

$$k = \max_i, \; r_{\text{eff}}(W_{:,2i-1:2i}),$$

In this case the max is essentially 1 across all pairs, suggesting that $k = 1$ is sufficient to capture these features without loss of information. (Table 2).

Mean	Std Dev	Range
1.0003	0.01	$[1.0, 1.05]$

Table 2: Effective rank of paired-feature encoder blocks. Values near 1 indicate collapse to a single direction.

5.2.1 Comparison of vanilla SPD and the softmax-and-attention extension

Both decompositions are trained with $C = 150$ subcomponents and the hyperparameters proposed in the original SPD paper, with the exception of the minimality loss, which is set to x for vanilla SPD and y for the softmax-and-attention extension. We use $k = 1$, since each pair was shown to be representable along a single direction in hidden space. The ground truth decomposition is one where each pair is captured by a single subcomponent.

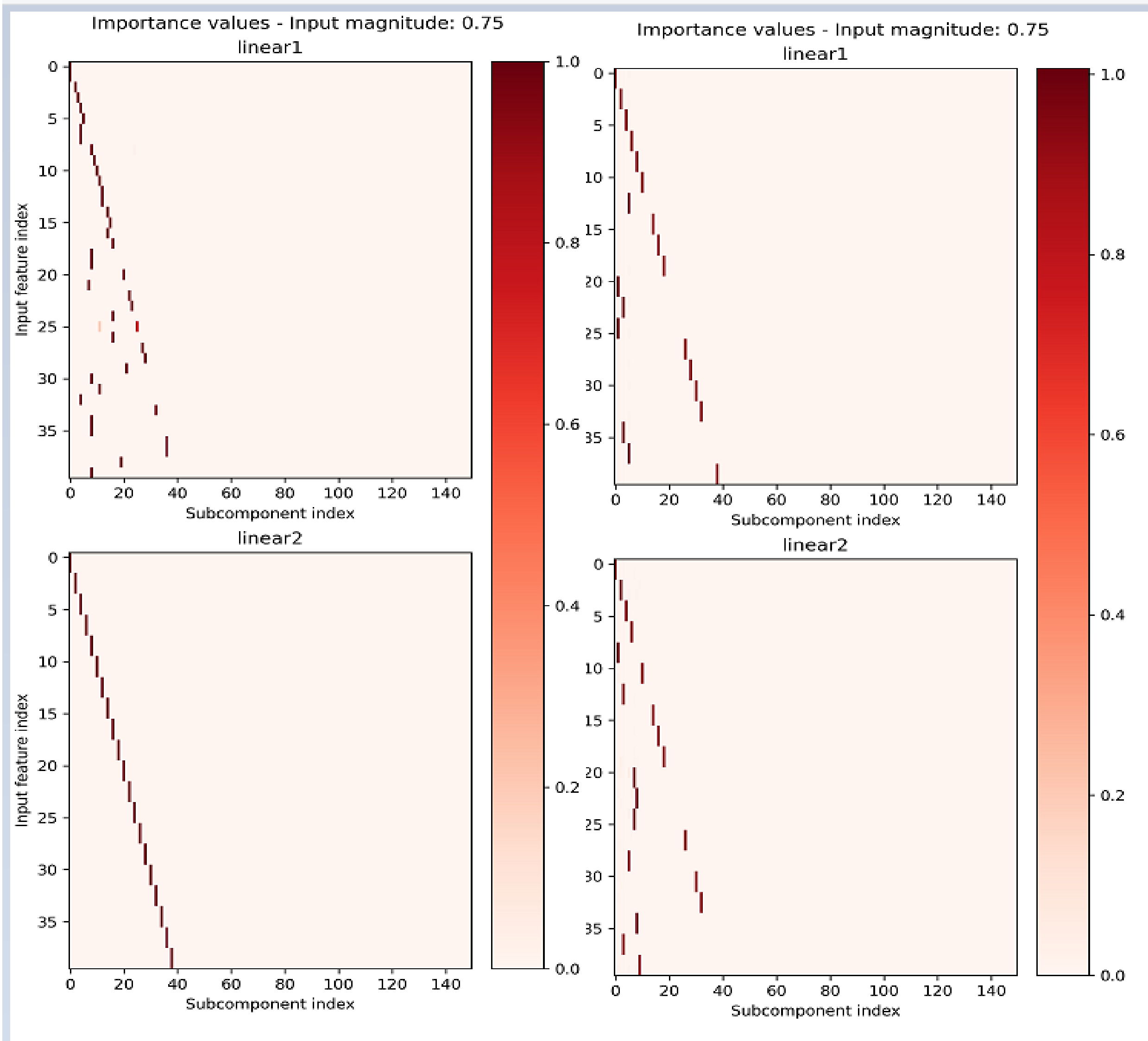

Figure 1: Left: Attention and softmax only at like 15k steps vs 40k steps, Right: Vanilla spd.

Text explaining what we see, this will change so wait for now. -Something about if both decompositions are faithful, and if they are faithful to one of the two hypotheses, if they are delta orthogonal, and if the importance values are interpretable.

To verify a successful decomposition, each matrix V_i should also be δ-orthogonal to the others.

The multi-dimensional superposition hypothesis predicts that the projection matrices for different irreducible features are pairwise δ-orthogonal. At the parameter level, the recovered V_c matrices are the operational counterpart, and the prediction translates to the requirement that recovered atoms have small pairwise overlap in their input-side directions.

For each method, we compute the pairwise cosine similarity [5]

$$M_{cc'} = \frac{V_c^\top V_{c'}}{\|V_c\| \|V_{c'}\|} \tag{9}$$

[5] Maybe even UV combined? Only read or write?

compared against the across-group cosine similarity of the trained model's encoder (0.053, 1), the softmax-and-attention shows "something" and the baseline model finds that "something else".

here we have four nice plots, overall alligment, across V's, across U's, entire thing? Reference to original model and maybe a table?

Remarks for this section Intra cluster cosine has a -1. Two features are perfectly alligned to a factor of -1, can this cause issues with decomposition? Same component could capture both features, if they read from the same input. However they shouldnt as its distinctly different feature indices.

5.3 Extending to Rank-k: When Higher-Dimensional Structure Matters

The rank-k extension is required in settings where feature representations do not collapse to a single direction. This is illustrated by a model trained to compute the volume of an n-dimensional simplex.

In this experiment, consider a model trained to compute the volume of an n-dimensional simplex:

$$f(x) = W_1\sigma(W_2x), \quad W_2 \in \mathbb{R}^{k\cdot n(n+1)\times d_{hidden}}, \quad W_1 \in \mathbb{R}^{d_{hidden}\times k}$$

Here, k denotes the number of simplices and n the dimensionality of each simplex, which has n+1 vertices.

As in Section 5.2, the k simplices are sampled independently with fixed probability, such that either all n(n+1) features of a simplex are active or none are. The resulting feature groups are therefore statistically dependent, and their joint distribution assigns negligible probability to partial activation patterns. By the definition of irreducibility in Section 4.2, each simplex constitutes an irreducible feature with formal dimensionality $d_{fi} = n(n+1)$.

For this experiment, the model is trained with $n = 2$ and $k = 4$, yielding four triangles $\mathbb{R}^2$ and a total of 24 input features. The ground-truth decomposition therefore consists of four subcomponents, each responding to 6 adjacent features. We note that the effective rank and cosine similarity here imply a decomposition should require at least $k = 2$. **[Placeholder: cosine similarity results]**

[Placeholder: effective rank results, showing $r_{eff} > 1$] Firstly, we demonstrate that the original SPD formulation surprisingly yields the best clustered reconstruction when $k = 1$:

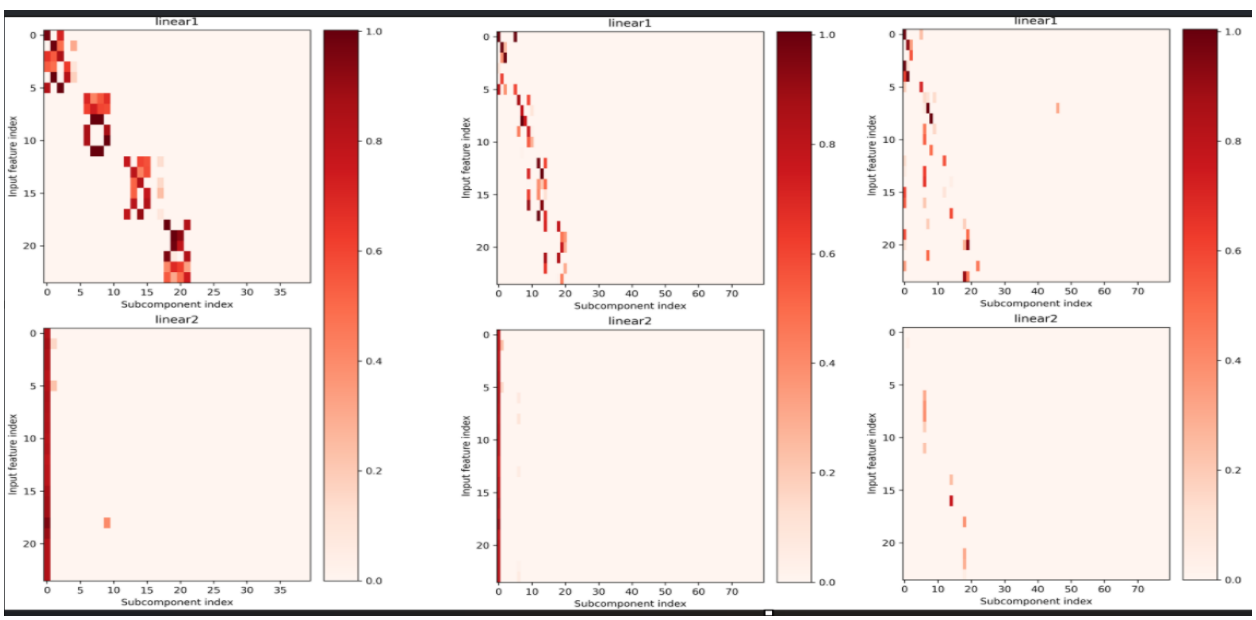

Figure 2: SPD decomposed with k=1,2,3 from left to right.

The reconstruction clearly yields 4 subcomponents per group, that only respond to the 6 features of a single simplex, however these subcomponents are seperated even though they are irreducible.

We note that we werent able to find a reconstruction at all when the minimality coefficient was outside of $[1e{-}4, 5e{-}4]$.

The softmax and attention extension, on the other hand, shows the expected behaviour, with $k = 1$, the reconstruction neccessitates that each simplex is split into multiple subcomponents, here each subcomponent captures a single feature. The extension to $k = 2$ allows the model to capture each simplex as a single unit, with all 6 features of a simplex captured by a single subcomponent. However, it does collapse 2 simplices into a single subcomponent. The extension to $k = 3$ does not yield any improvement, suggesting that the model has found a faithful reconstruction at $k = 2$.

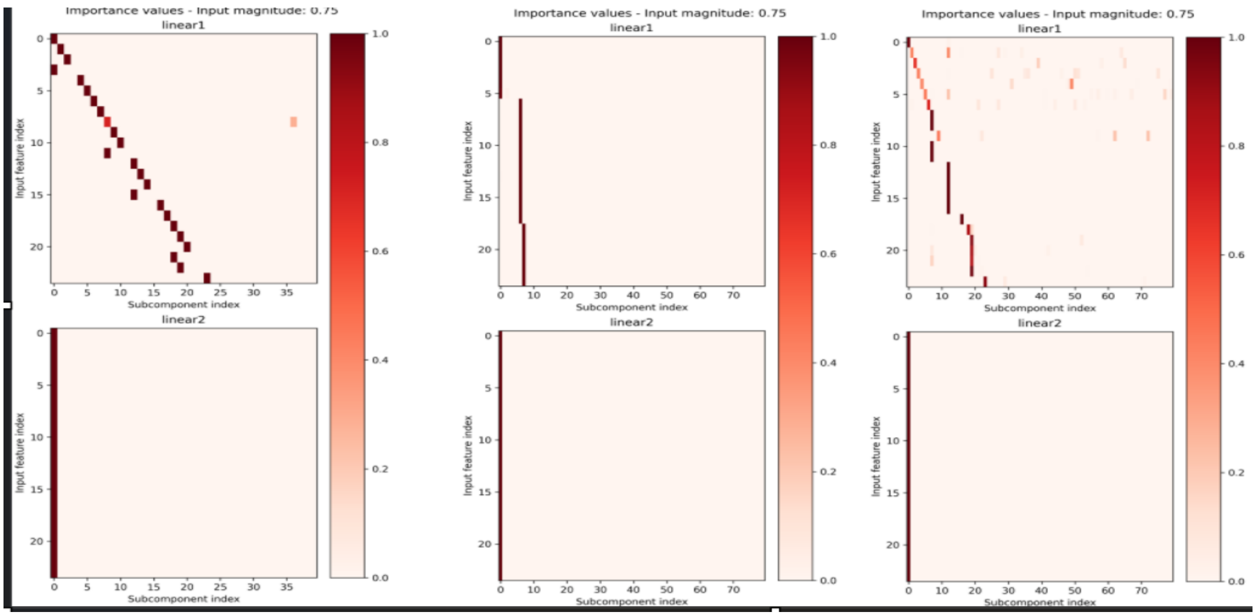

Figure 3: Softmax-and-Attention decomposed with k=1,2,3 from left to right.

While the softmax approach does not find a perfect reconstruction either, it clearly demonstrates how the extension from rank 1 to rank 2 allows a non diagonal representation of the features. Had it been perfect, then the middle plot would have shown 4 components instead of the collapsed 2.

Same plots i was going to do for the TMS experiment, perhaps this gets its own section, we compare how input spaces are alligned over training, does the softmax encourage δ orthogonality better?

5.4 Checking for orthogonality throughout training with softmax and attention / compare to baseline spd.

5.5 Toy model of learning gates

Remark: Im not sure we can sample binary values, double check the distribution, i think it requires a continuous thing with b not fixed? Could always train an addition model, but they already did that in the original paper? Can we just use that one to prove the claim?

To test the case of enges, where the probability distribution is in fact reducable. We could train it to learn logic gates i.e 1+0 = 0, 1 + 1 = 1, 0 + 0 = 0, 0 + 1 = 1, in such a view each could occur independently ,but nlywhen both are active, the output is active, we then sample independently from the two features and see if it compacts them in a single unit or not.

The concept here is clear: The ideal decomposition here is one that produces independent features, that can co occur, however the atomic solution is independence.

We also need to probe for a paired result here. The ground truth should show part ownership across 2 components.

bool A, bool B are independnet their joint result shoul be shared ownership.

5.6 Extending the rank to be an upperbound through per feature ablation

5.7 Irreducibility as the target of decomposition

6 Discussion

7 Acknowledgements

8 References

9 Other things to consider?

Minimal descriptive length?

9.1 That some features are multidimensional

Think back to SAE, that learn a dictionary of features, then find hte linear combination of those features, however that this fails for cyclical features, as its cheaper to learn a single feature, than learn a direction and a periodic function. Now for your experiment, the dream scenario is that we allowe a multi dimensional component, and that this can learn the cyclical feature, i.e a weekly cycle or modular arithmatic either do this or extend TMS to be multi-dimensional, show if it works better with ur gnn or not, likewise we need to show that through individual masking of features, we can teach a component to kill one of its features, if it is better explained by a rank 1 component than rank 2.

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9.1.1 GNAN-based importance functions

In SPD, each importance function γ_c^l is an independent MLP that receives only its own inner activation. Intuitively, however, causal importance is relational: whether a subcomponent can be ablated without affecting the output may well depend on which other subcomponents are active alongside it.

To address this, the independent MLPs are replaced with a Graph Neural Additive Network (GNAN) that can relate each subcomponent to its neighbourhood.

The C subcomponents are treated as nodes in a graph, with node features given by their inner activation vectors $\mathbf{h}_c^l(x) \in \mathbb{R}^k$. The GNAN updates each node's representation through an additive neighbourhood aggregation. For node i in layer l and feature dimension k [6]:

$$[\mathbf{z}_i]_k = \sum_{j=1}^{N} \frac{1}{|\mathcal{L}(j)|} \cdot \rho\left(\frac{1}{1+\mathrm{dist}(j,i)},\ \cos(j,i)\right) \cdot f_k([\mathbf{h}_j]_k)$$

OBS: Implementation ill present today is this:

[6] I am very open to alternative equations, i.e perhaps we delete the layer distance inside rho, however it seemed necessary to use cosine similarity, as otherwise the signal was useless as described in some earlier section

$$\mathbf{z}_i = \sum_{j=1}^{N} \rho\left(\text{layer distance}, \text{cosine similarity}\right) \cdot f_k\left([\mathbf{h}_j]_k\right)$$

where f_k is a learned function applied to the k-th inner activation of each neighbouring node j; $|\mathcal{L}(j)|$ is the number of nodes in the same layer as j, which normalises the contribution of each layer; $\text{dist}(j, i)$ is the layer distance between nodes j and i[7]; $\cos(j, i)$ is the cosine similarity between their parameter vectors; and ρ is a learned function that maps both the distance kernel and the cosine similarity to a scalar weight.

z_i becomes a vector in $\mathbb{R}^k$, which fits perfectly with the current activation vectors.

A separate MLP is trained to combine the aggregated neighbourhood information and the node's own activation into an importance value in $[0, 1]$: Producing a single importance value per subcomponent and passing it through a leaky hard sigmoid, following SPD [BBS25]:

$$g_c^l = \sigma_H(AGG(\mathbf{z}_i^l, \mathbf{h}_i^l))$$

The self-connection through f_{self} (or just the raw activation) ensures that each node retains its own identity after aggregation, mitigating the oversmoothing that can occur in GNNs. {Clever cite here, i.e gnn book}

Alternative formulation, that would work with the next section

After the GNAN update, the per-dimension importance values are computed from the updated representation and passed through a leaky hard sigmoid, following SPD [BBS25]:

$$\mathbf{g}_c^l(x) = \sigma_H\left(\mathbf{h}_c^l(x)\right) \in [0, 1]^k$$

in other words, the importance function becomes multidimensional, and each column can be masked independently.

Remarks:
For now the edge set construction is left as the fully connected graph, which remains reasonably computationally expensive for small models. However it could very easily be made more efficient, for other architectures, or by considering only the k-hop neighbourhood of each node, or only considering prior neighbours or later neighbourhoods. Likewise we could store both distances as sparse structures as they are symmetric. $\cos(j, i) = \cos(i, j)$

Remark 2:
The choice of edge set construction and distance function is a very important design choice, e.g using the layer distance on a fully connected graph encodes the same signal for all nods in a layer: For nodes i_1, i_2 in the same layer l:

$$\forall\, j: \quad \text{dist}(j, i_1) = \text{dist}(j, i_2) = |k_j - l|$$

and therefore

$$[\mathbf{h}_{i_1}]_k = \sum_{j=1}^{N} \frac{1}{\#\text{dist}_{i_1}(j, i_1)} \cdot \rho\left(\frac{1}{1 + |k_j - l|}\right) \cdot f_k([\mathbf{x}_j]_k) = [\mathbf{h}_{i_2}]_k$$

since no term in the sum depends on the identity of i within layer l, only on the layer index l itself.
Additionally this produces a counterproductive signal, the gnan tries to boost or suppress all nodes in a layer together, instead of learning to differentiate between them.

Remark 3:
The choice of distance should probably be reconsidered in general, I have a feeling that the function needs to be from an asymmetric metric space, i.e

$$dist(i, j) \neq dist(j, i)$$

Atleast if the intution is if component A in layer k, is highly active, perhaps component B in layer k+1 should likely be inactive, as the other component needs to handle this feature, thus the distance from A to B should be different, as otherwise both are boosted or suppressed together.

Remark 4:
The current experiments are more sensitive to hyperparameters, than the original code seems to be, if the minimality pressure is too low, then all components are active, and doesn't "kill" any of them

Remark 5:
I haven't been able to find any "benefit" in setting k higher, this is ofcourse because hte ground truth of all the models

[7] The layer distance, could later be signed to differentiate between previous and future layers

we have are rank 1 independent features, we need to find a scenario where the dependence and or rank k is actually necessary:

My ideas

- Sample points from the unit circle, i.e $\cos(\theta)^2 + \sin(\theta)^2 = 1$ then
$$x = \sqrt{1 - \sin(\theta)^2} \text{ or} - \sqrt{1 - \sin(\theta)^2}$$
i.e sometihng can only be expressed with a rank 2 component?
- we could look for circuits, the goal is then no longer to find N independent features, but suppose we have input groups [0,1,2], [2,3,4], [3,4,5] then we want one component that activates for the first group, one for the second and one for the third

Remark 6:
I added the gnan plots to wandB remember to show them

Remark 7:
The per feature construction with only 1 feature is rather boring, and does not utilize the expressiveness at all.

Remark 8:
I dont think its a code problem, as i can reproduce equivalent results to their findings. Im just not able to beat them. We need to discuss experiments where the k dimensionality and dependence actually matters.

10 Notes

Putting the minimality lsos entropy higher, encourages diagonals? Instead of capturing two features in the same component.